

Week 4

Reducibility

m-reducibility (many-one reducibility)

Definition (m-reducibility):

Let $A \subseteq \mathbb{N}$, $B \subseteq \mathbb{N}$. We will say that A is m-reducible to B (and write $A \leq_m B$) if there exists a computable function $f: \mathbb{N} \rightarrow \mathbb{N}$ such that,

$$\begin{aligned} &\text{for every } x \in \mathbb{N}, \\ &x \in A \iff f(x) \in B. \end{aligned}$$

-Notice that the definition does not require any of the sets to be decidable, or semi-decidable.

- Notice that the domain $\mathbb{N}$ can be replaced by Σ_1^* for some finite alphabet Σ_1

and the co-domain $\mathbb{N}$ can be replaced by Σ_2^* for some finite alphabet Σ_2

Recall: For any finite alphabet Σ , there exists a computable bijective function from Σ^* to $\mathbb{N}$ (by means of Godel numbering or an equivalent).

Activity 1

Let A, B be subsets of $\mathbb{N}$.

Let $A \oplus B = \{2x : x \in A\} \cup \{2x + 1 : x \in B\}$.

$A \oplus B$ is read as A join B

Prove that both A and B are m-reducible to $A \oplus B$.

Proof:

We start by proving that $A \leq_m A \oplus B$.

We need to show that **there exists** a computable function $f_A: \mathbb{N} \rightarrow \mathbb{N}$ such that for every $x \in \mathbb{N}$,

$$x \in A \iff f_A(x) \in A \oplus B.$$

Let f_A be given by $f_A(x) = 2x$. Clearly f_A is computable (by CT).

Remains to show that, for an arbitrary natural x , $x \in A \iff 2x \in A \oplus B$.

Activity 1 continued

Proving that $x \in A \iff 2x \in A \oplus B$ (for an arbitrary natural x)

$\Rightarrow$:

$$\begin{aligned}x \in A &\Rightarrow 2x \in \{2n : n \in A\} \\&\Rightarrow 2x \in \{2n : n \in A\} \cup \{2n + 1 : n \in B\} = A \oplus B.\end{aligned}$$

$\Leftarrow$:

$$\begin{aligned}2x \in A \oplus B &\Rightarrow 2x \in \{2n : n \in A\} \cup \{2n + 1 : n \in B\} \\&\Rightarrow 2x \in \{2n : n \in A\} \vee 2x \in \{2n + 1 : n \in B\} \\&\Rightarrow 2x \in \{2n : n \in A\} \vee \text{FALSE} \\&\Rightarrow 2x \in \{2n : n \in A\} \\&\Rightarrow 2x = 2n \text{ for some } n \in A \\&\Rightarrow x = n \text{ for some } n \in A \\&\Rightarrow x \in A\end{aligned}$$

m-reducibility (many-one reducibility)

Let $A \subseteq \mathbb{N}$. We will say that A is m-reducible to $A \oplus B$ (and write $A \leq_m A \oplus B$) if

there exists a computable function $f: \mathbb{N} \rightarrow \mathbb{N}$ such that,

for every $x \in \mathbb{N}$,

$$x \in A \iff f(x) \in A \oplus B.$$

Activity 2

Let A be a decidable subset of $\mathbb{N}$. Let B be a subset of $\mathbb{N}$.

Prove that, if $B \leq_m A$, then B must be decidable as well.

Proof:

From the given:

- 1) we can compute the characteristic function of A (let us call it I_A).
- 2) There exists a computable function $f: \mathbb{N} \rightarrow \mathbb{N}$ such that, for every $x \in \mathbb{N}$,

$$x \in B \iff f(x) \in A.$$

Now we want to show that the characteristic function of B (let us call it I_B) is computable.

We will use the Church-Turing thesis.

Activity 2 contiued

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def I_B(x):  
    return I_A (f (x))
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By CT, I_B is a computable function.

This means B is decidable.

Homework

Solve the previous activity with “semi-decidable” instead of “decidable”.

Activity 3

Recall that K is the Kleene halting set, and H is its sister (check last lecture).

Prove that $K \leq_m H$.

Proof:

Recall: $K = \{e : M_e(e) \downarrow\}$, $H = \{(e, x) : M_e(x) \downarrow\}$.

Now we want to prove that **there exists** a computable function

$f: \mathbb{N} \rightarrow \mathbb{N}^2$ such that, for all $x \in \mathbb{N}$, $x \in K \iff f(x) \in H$.

Indeed, f given by $f(x) = (x, x)$ is an m-reduction of K to H .

More clearly, for an arbitrary natural x , $x \in K \iff (x, x) \in H$.

Do you see that this equivalence is true? If not, try to prove it

We want to show that $x \in K \iff (x, x) \in H$ (for an arbitrary natural x)

$$x \in K \iff x \in \{e : M_e(e) \downarrow\} \iff M_x(x) \downarrow \iff (x, x) \in H$$

Activity 3.5

Let A be a decidable subset of $\mathbb{N}$. Let B be a subset of $\mathbb{N}$.
Suppose that $A \leq_m B$. Can we conclude that B is decidable as well?

First guess: NO (healthy)

No because for some output in B we might be searching forever for an input in A if it doesn't exist. The "decider" for B is not guaranteed to halt on $f(x)$ (where f is the reduction $A \leq_m B$).

Suppose now we want to prove that such a conclusion is false.

We need a counter-example. We need sets X, Y of natural numbers such that X is decidable, X reducible to Y , but Y is not decidable.

Take this: $X = \mathbb{N}$, $Y = K$, and let $k \in K$. Consider now the function $f: \mathbb{N} \rightarrow \mathbb{N}$ such that $f(x) = k$. Clearly f is computable and satisfies

$$x \in X \Leftrightarrow x \in \mathbb{N} \Leftrightarrow \text{TRUE} \Leftrightarrow k \in K \Leftrightarrow f(x) \in K$$

What did we mean by a counter-example?

The statement discussed on the previous slide is a universal statement (Starts with for all, implicitly)

Here is the statement for clarity:

Let A be a decidable subset of $\mathbb{N}$. Let B be a subset of $\mathbb{N}$.

Suppose that $A \leq_m B$. Then B is decidable as well.

Which is equivalent to:

For all A, B subsets of $\mathbb{N}$ ($\forall A, B \in P(\mathbb{N})$),

if

$A \leq_m B$ and A is decidable,

then

B is decidable.

The statement is a universally quantified implication

$$\forall A, B \in P(\mathbb{N}), \\ (A \leq_m B \wedge A \text{ is decidable}) \Rightarrow B \text{ is decidable}.$$

A counter-example to a universal statement is a case in which the statement fails. In our context here, a counter example will be an example of sets A, B for which $(A \leq_m B \wedge A \text{ is decidable})$ is true while “ B is decidable” is false.

Activity 4

Prove that every decidable set is m-reducible to $\{1\}$.

Homework

Let X be a proper subset of $\mathbb{N}$ (i.e. $X \subsetneq \mathbb{N}$). Prove that $\emptyset \leq_m X$. You may assume that the universes are both $\mathbb{N}$.

Hint: Consider two cases, X is empty and X is nonempty

1-reducibility (denoted $\leq_1$)

This is just an m-reducibility with the extra requirement that the reduction function is injective.

For example, we have $K \leq_1 H$. *Any other examples or non-examples from the activities we solved?*

1-reducibility is **stronger** than m-reducibility

Activity 5

A set A of natural numbers is a *cylinder* if there is a set B of natural numbers such that $A \equiv_1 B \times \mathbb{N}$.

Prove that: S is a cylinder iff for any set C , $C \leq_m S \implies C \leq_1 S$.

The $\equiv_1$ means $\leq_1$ in both directions.

Turing Reducibility (denoted $\leq_T$)

Observe how m-reducibility was defined very naturally for sets.

Now we define a concept of reducibility between functions

Definition (Turing reducibility):

We will say that f is Turing-reducible to g (and write $f \leq_T g$) if we can write a program that uses g to compute f .

I.e., if we can compute g , then we can compute f .

It is common to say f is g -computable to mean that $f \leq_T g$

Turing Reducibility for sets

For sets, we write $A \leq_T B$ to mean that $I_A \leq_T I_B$.

We also say A is B -decidable (if B is decidable, then “ B -decidable” is the same as just “decidable”).

Intuition for Turing reducibility

Imagine you are writing a program to compute f , and that program is making use of g (as a black box). This is exactly what $f \leq_T g$ means.

It is like saying, f is computable (modulo g). So f looks computable except that the program uses g for some inner working

Observe that f, g are not required to be computable for Turing reducibility to make sense. In fact, the concept of Turing reducibility is interesting particularly when these functions are not computable

Activity 6

Prove that $\bar{K} \leq_T K$ but not $\bar{K} \leq_m K$.

Activity 7

Prove that, for every decidable set D , we have $D \leq_T \emptyset$

Remark (think about it)

For any two sets A, B we have that

$$A \leq_1 B \implies A \leq_m B \implies A \leq_T B$$